

Water Containers Queries

Time limit: 1.00 s

Memory limit: 512 MB

There are two water containers: container A has volume a and container B has volume b . You want to measure x units of water using the containers.

Initially both containers are empty. On each move, you can fill a container, empty a container or move water from a container to another container. When you move water, you must always fill or empty at least one container. After the moves, container A must have x units of water.

Your task is to efficiently check if it is possible to measure the water in several cases.

Input

The first line has integer n : the number of tests.

After this, there are n lines. Each line has three integers a , b and x .

Output

For each test, print YES if it is possible to measure the water and NO otherwise.

Constraints

- $1 \leq n \leq 1000$
- $1 \leq a, b, x \leq 10^9$

Example

Input	Output
7	YES
5 3 4	YES
1 1 1	NO
1 1 2	NO
2 2 1	YES
123 456 42	YES
1000 999 123	NO
1000 998 123	